

CoopGBFS: A Federated Learning and Game-Theoretic-Based Approach for Personalized Security, Recommendation in 5G Beyond IoT Environments for Consumer Electronics

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Abstract—In 5G IoT for consumer electronics security recommendation systems, Machine Learning (ML) based Federated Learning (FL) methods plays an important for the detection of cyberattacks. For this objective, several different models for effective feature selection and recommendation systems are presented in the literature. However, presented models prone to misclassify 5G Internet of Things (IoT) malicious traffic due to inappropriate feature recommendation system. To address the issue, In this paper, firstly, an Automatic Data set Generator (ADG) method is proposed and then designed ADG algorithm for the effective features set dataset. Then based on proposed ADG algorithm we proposed a new recommendation model CoopGBFS, based on cooperative game theory, federated learning and correlations with algorithm accuracy to address the problem. Finally, based on proposed CoopGBFS method an algorithm named CoopGBFS is developed and designed to select and recommend effective features set for 5G IoT network security. For the proposed approach evaluation, Bot-IoT dataset and four well-known ML algorithms are utilized. However, from the result analysis, it is cleared that the presented method is efficient for 5G IoT malicious traffic detection features recommendation for Consumer Electronics.

Index Terms—CoopGBFS, recommendation system, 5G IoT network, cyberattacks detection, federated learning, for consumer electronics.

I. INTRODUCTION

RECENTLY, 5G IoT technology received much importance, specially in Consumer Electronics, and

its growth is increasing every minute due to its efficient performance results. Moreover, by introducing this technology, life becomes very comfortable due to easy accessibility. In the initial stage of IoT technology Consumer Electronics, the administrator can only install it in small offices and homes, then gradually, this technology extended and grew up very rapidly. Then to save time and control everything efficiently, the technology is integrated into industries for more reliability. Now the result is that IoT in consumer electronics has become a very important part of our life [1]. At the end of this year, the interconnected IoT network consumer electronics will be expected 27.1 billion, which is expected that globally, 3.5 IoT consumer electronics devices per person. However, this is a profound change in IoT consumer electronics technology. It is fascinating that this technology is expending very fast, but due to this development, cyber-attacks are also gaining popularity and becoming an emerging issue. Nevertheless, to overcome cyber-attacks problems and protect their personal information and unauthorized access, several researchers in the research community endeavor hard and present several different efficient models. However, for this purpose, the researchers try and present effective Machine Learning (ML) based cybersecurity systems models on overcoming rising cyber-attacks issues in 5G IoT network security [2], [3].

In 2017, Denial of Service (DDoS) malicious attacks became the most spreading and growing attacks and got much attention [4]. Similarly, in a report issued by Kaspersky Lab, the cyber-attacks in IoT networks in 2017 is much increased compared to cyber-attacks in 2013. However, most attacks were in 2017 are DDoS and Botnet attacks, which is very harmful compared to other cyber-attacks. However, to overcome these attacks, an effective intrusion detection system is necessary to develop. For this reason, to address the above problem, Anderson 1980 design and presented the first Intrusion Detection System (IDS) [5]. Similarly, in 1987, D. E. Denning introduces a new intrusion detection model capable of detecting harmful intrusion. However, their model is based on a hypothesis and real-time intrusion detection, the primary purpose of his proposed system is identifying malicious traffic. Nonetheless, recently the most rising security

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attacks are man-in-the-middle harmful threats with (DDoS) attacks [6], [7]

Recently, in our previous work [3], [8], we studied and addressed the problem of effective feature selection and presented numerous effective feature selection approaches. Furthermore, we used well-known machine learning (ML) algorithms to overcome feature selection and identify cyber-attack traffic flows. However, in our studies, we showed that selection of more features could lead to computational complexity and affect the performance of the proposed approach or accuracy results. For instance, the selection of more than ten feature sets can slow the presented model or slow the ML algorithms process. Hence, the selection of efficient features for the identification of malicious attack traffic in 5G IoT consumer electronics network environments using machine learning algorithms and the federated learning technique is a crucial and emerging issue [9].

To overcome the problem of effective feature selection and recommendation for identifying cyber-attacks in IoT networks, in this paper, we proposed a new framework and algorithms by using well-known machine learning algorithms and an open-access dataset in order to achieve improved performance outcomes. Nonetheless, the key contributions are discussed in the paper are:

- To tackle the issues associated with feature selection and security recommendation system problems in 5G IoT network for consumer electronics malicious traffic flow. In this manuscript, we introduce a novel framework model named CoopGBFS based on cooperative game theory to accurately deal with the above problem and achieve promising performance results.
- Then based on the proposed method, correlations with algorithm accuracy are applied to address the problem accurately. Firstly, a novel feature ranking method correlation is applied for the ranking of features, and then based on correlation, an algorithm is applied in order to achieve improved performance outcomes in 5G IoT networks.
- Subsequently, the cooperative Shapley value is employed by leveraging cooperative game theory. A considerable number of scholars within the academic research community have employed game theory as a means to address decision-making challenges, yielding highly favourable outcomes. Therefore, cooperative game theory was also employed to rank the features in the context of detecting cyber-attacks in the 5G IoT network, utilising machine learning algorithms.
- Afterwards, leveraging the cooperative shapely value of game theory and correlation, researchers have devised an algorithm called CoopGBFS. This algorithm aims to effectively determine the optimal feature set for identifying IoT network traffic. The algorithm being proposed is based on the wrapper technique, which is employed to prioritise the most effective feature. This technique offers an adequate quantity of data for the detection of malicious network traffic within the framework of the Internet of Things (IoT). The methodology proposed in this study entails the utilisation of the Bot-IoT dataset and

the evaluation of four well-established machine learning algorithms.

- In the end, we reached a conclusion and subsequently presented a set of promising features that possess sufficient information to effectively detect cyber-attacks in IoT consumer electronics network environments. These features were identified using a technique grounded in cooperative game theory. Based on the examination of the empirical findings, it is evident that the chosen set of features possesses discriminative capability and contains sufficient information to effectively detect instances of cyber-attacks within IoT networks.

The paper is structured into the subsequent sections: The subsequent section provides an introduction to the works that have been reviewed. Next, the present study is exemplified in Section III. Section IV delves into the discourse surrounding the evaluation methods. Section IV of the document delves into the examination and evaluation of the subject matter, encompassing both discourse and analysis. Section V encompasses the presentation of the conclusion and future work.

II. RELATED WORKS

This section aims to examine and highlight the existing literature on the detection of cyber-attacks or malicious traffic flow in the network environment of IoT consumer electronics. In our previous research conducted in [3], we presented several different effective models for effective feature selection and effective network traffic detection, such as IM flow classification and malicious Bot-IoT attacks detection. Nonetheless, it is proved in several studies that feature selection is an fundamental step of ML [10]. The primary objective of efficient feature selection is to identify a set of features that possess discriminative capabilities in detecting cyber-attacks, while simultaneously eliminating redundant features from a larger pool for utilisation in machine learning algorithms. Although redundant features are a sign of features that are insufficiently informative for the detection of cyber-attacks in IoT consumer electronics network environments, they are also a sign of features that are failing to meet the needs of the IoT industry. In order to address the challenge of efficient feature selection, numerous researchers have made significant efforts and put forth effective models, such as: in 2018 Egea et al. in [11] studied and reviewed several proposed effective feature selection techniques. The techniques that they reviewed in their studies are based on correlation measurement. Based on the reviews technique, they introduced a new technique called Fast Based Correlation Features (FCBF) to enhance the performance and select effective features in the industrial environment. After that, the researchers made a modification to the initially proposed technique, which resulted in the introduction of the FCBFiP technique. Similarly, Meidan et al. (2018) [12] conducted a study on the detection of cyber-attacks in the context of IoT networks. Their research focused on the identification of malicious attacks in 5G IoT networks that originate from IoT devices. In their study, they empirically evaluate their proposed adversarial cyber-attack detection method.

Algorithm 1: automatic dataset generator:**Input:**

- x : matrix of size $m \times n$ with m samples and n features
- p : probability of selecting a feature in the subset generation phase
- important_samples : set of indices for important samples
- α : threshold for feature values to be considered as having desired information

Output: -recommended set: // set of indices for chosen features**1. Begin****2. // Subset generation phase**

```

3. for (int i = 1; i <= n; ++i) { // Loop over all features
4.   if (rand() < p) { // Randomly select feature with probability p
5.     subset.insert(i); // Add selected feature to subset
6.   }
7. }
8. for (int j = 1; j <= m; ++j) { // Loop over all samples
9.   float score = 0.0; // Initialize score for this sample
10.  for (auto i : subset) { // Loop over selected features
11.    score += x[j][i]; // Add feature value to score
12.  }
13.  scores[j] = score; // Store score for this sample
14. }
15. float threshold = median(scores); // Compute median score
16. for (auto i : subset) { // Loop over selected features
17.   for (auto j : important_samples) { // Loop over important samples
18.     float mean_i = mean(x, i); // Compute mean feature value
19.     if (mean_i > threshold) { // Accept feature if mean > threshold
20.       accepted.insert(i); // Add accepted feature to set
21.     }
22.   }
23. }
24. bool has_desired_info = false; // Assume feature does not have
25. // Validation of subset phase
26. for (auto i : accepted) { // Loop over accepted features
27.   if (x[j][i] > alpha) { // Check if feature value is above threshold
28.     has_desired_info = true; // Feature has desired information
29.     break; // Stop searching for this feature
30.   }
31. }
32. if (has_desired_info) { // Feature has desired information
33.   chosen.insert(i); // Add chosen feature to set
34. }
35. // Output chosen features
36. for (auto i : chosen) { // Loop over chosen features
37.   std::cout << "Chosen feature " << GL("alpha")
38.   << " " << i << std::endl;
39. }
40. } end if, end for;
41. } End Return

```

Fig. 1. Proposed Dataset Generator Algorithm (DGA).

Similarly, several models in the research community presented many effective ML models for the identification and performance improvement and achieved effective results. Thus, from the above reviews related to robust feature for IoT traffic identification, It is evident that this topic can be thoroughly investigated, wherein effective features can be chosen to sufficiently convey information for the detection of detrimental environmental flows.

The proposed algorithm that generate data set for the proposed model is shown in Fig. 1 in detail. The 5G IoT Network and detection method is shown in Fig. 2 in details. While proposed framework is shown in Fig. 3

III. PROPOSED COOPGBFS METHOD

In this particular section, the presented framework CoopGBFS explains and tries to demonstrate the step-by-step process in detail to overcome the issue related to the selection of features for cyber-attacks 5G Internet of Things traffic identification in 5G IoT consumer electronics [13] network environments as shown in Fig. 3 The essential part of the feature selection process is to rank effective features that have more information for the harmful identification and remove a redundant feature. In order to achieve this objective, we initially employed a novel correlation technique to prioritise

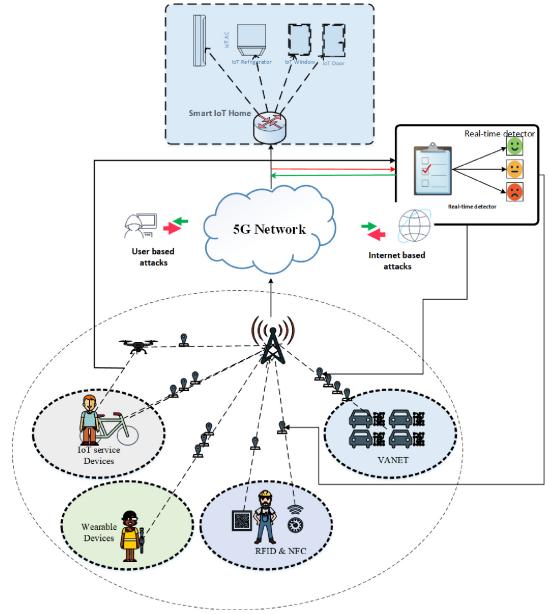


Fig. 2. Real Time Detector in 5G Based IoT Network.

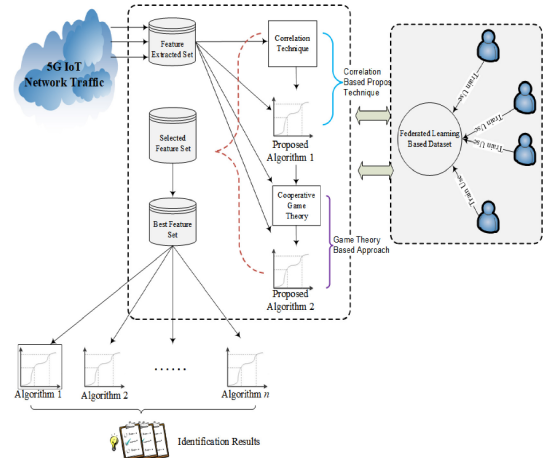


Fig. 3. Proposed CoopGBFS Recommendation Model.

the influential features and eliminate the superfluous features from the trace traffic dataset. Once the process of ranking and selecting the optimal feature set has been completed, a widely recognised algorithm is employed to determine the feature that provides the most informative content for the identification of malicious traffic. Then, based on the above-explained technique, the redundant features are removed, and effective features are selected. The primary purpose of applying the algorithm is to remove unwanted features that are not applicable or don't have discriminative power to identify malicious attacks in 5G IoT consumer electronics network environment.

Nevertheless, the utilisation of cooperative Shapley values has proven to be effective for the purpose of selection. In conclusion, the CoopGBFS algorithm is introduced as a means to determine a proficient set of features for the classification of network traffic in 5G IoT, utilising the cooperative Shapley value. The algorithm under consideration relies on the filter method to prioritise the efficient features that possess significant potential for identifying traffic in the context of 5G

network. The methodology proposed in this study entails the utilisation of the Bot-IoT dataset and the evaluation of four well-established machine learning algorithms. Nevertheless, the examination of the empirical findings illustrates that the suggested methodology proves to be efficacious in the detection of malevolent network activity within the realm of 5G Internet of Things (IoT). In order to facilitate comprehension and provide a systematic approach, the subsequent section elucidates the comprehensive methodology of the proposed framework.

A. Feature Selection

This section introduced the applied metrics are discussed in detail. The detailed metrics are discussed in the following subsection.

1) *Correlations Based Metric*: In this subsection, we applied the Person Moment Correlation method to solve the effective feature selection problem or rank the most effective features in the 5G IoT consumer electronics network environment. To remove redundant features and rank effective features correlation technique is the best choice. The technique is useful for the identification of the relationship between features and target features in a dataset. In 1889, R. E. Fancher proposed this technique [14]. After sixteen years, researcher K Pearson updated the technique with the name Person Product Moment Correlation. However, the main purpose of updating this technique is to identify the relationship among class attributes or features. The updated technique is based on statistical analysis, which ranks attributes of a class. Here attributes indicate the features of the class. For instance, let A and B features, then to find out the correlation between A and B, the following formula can be used.

$$C_{X,Y} = \frac{\text{Covariance}(A, B)}{\sigma_x \sigma_y} \quad (1)$$

$$C = \frac{\sum_{i=1}^n (a_i - \bar{a})(b_i - \bar{b})}{\sqrt{\sum_{i=1}^n (a_i - \bar{a})^2} \sqrt{\sum_{i=1}^n (b_i - \bar{b})^2}} \quad (2)$$

In equation (1), CA, B indicates the relationship among attributes or simply the correlation coefficient, While covariance is (A, B). Whereas ab indicates the standard deviation for features A and B. However, the above explanation is for a simple correlation relationship, while Eq. (2). can be used for two set attributes correlation coefficient evaluation. For the two sets of attributes correlations coefficient, we used Eq. (2). For instance, if A and B are sets of features, then a1, b2, c3, then the correlation among features is very strong. Likewise, if the value of the C coefficient is minus one (−1), the correlation between features is weak, not strong. While if there is zero value of C, then there is no relationship between features of classes. Thus, it is clear from the explanation of the correlation coefficient that this technique is useful for the feature selection and ranking in the high dimensional dataset. Thus, the effectiveness of the feature of classes can be analyzed by using Eq. (3) as follows:

$$\text{Corr} = \frac{\text{kavg}(\text{corr}_{fc})}{\sqrt{k + k(k-1)\text{avg}(\text{corr}_{ff})}} \quad (3)$$

In Eq. (3), Corr indicates the correlation coefficient of the final value and the relationship between features of classes. At the same time, Kavg(corr_{fc}) is the average of correlations of features. Similarly, K indicates the number of features set and while Avg(corr_{ff}) refers to average relationships among class features. As discussed in the previous section that applying the correlation coefficient technique is essential for effective feature selection. However, here are some factors before using the correlation coefficient technique for the effective feature selection for malicious IoT network traffic flow identification using the machine learning method. If the correlation between features is strong and the correlation coefficient between features and classes is weak, then it's mean the correlation is weak. Likewise, if the correlation between the feature and classes is strong, then it's known as strong correlations.

2) *Accuracy (ACC) Wrapper Based Metric*: As discussed in the previous subsection, the most critical part of the identification process is selecting effective IoT malicious traffic identification features. Unfortunately, it is challenging for a network administrator to determine the outmost feature for Machine Learning (ML) classifiers. To overcome this problem, we used the wrapper technique based on algorithm accuracy (ACC). The accuracy (ACC) wrapper technique effectively ranks effective features when there is a high-dimensional dataset. As we are interested in ranking, the feature gives enough information to identify cyber-attacks traffics in IoT networks. Thus we adopted the accuracy (ACC) wrapper technique for this study. The highest values of the ACC metric indicate the feature is robust and could achieve effective performance results. On the other hand, if the ACC metric value is not high, then the classification performance results will not be effective. Thus, applying the ACC wrapper-based metric is vital for ranking features in order to achieve improved performance outcomes in IoT network malicious traffic identification using ML algorithms.

3) *Cooperative Game Theory*: Game theory is a branch of mathematics and is widely applied in the economic field due to better performance results [15]. Due to game theory, a competitive situation can be understood in which decision-makers interact. Therefore, game theory has been applied in many areas such as economics, war, computer science, and business [16]. However, recently game theory has become scorching top in the field of computer science. In cooperative game theory, the main and most necessary part is the coalition. "Coalitions are the set of players as $N = 1, 2, 3, \dots, n$ and the coalition is a subset of players, N used for a binding agreement. Similarly, any subset of N , including N itself, can form a coalition [17]. More in-depth, a coalition game is a pair of (N, V) , where N is a finite set of players, indexed by $1: 2 \dots n$ arrow R associate with each coalition K subset N a real-valued payoff $v(k)$ satisfying $v(0) = 0$ ".

Recently in game theory, Shapley's value got much attention, specifically in computer science. Shapley first introduced the Shapley value. Then, due to its promising results for coalitions in game theory, the technique got much attention and adopted. The cooperative Shapley value is very effective for the ranking problem. This paper adopted cooperative Shapley value in game theory to overcome effective features selection

cyber-attacks traffic in IoT consumer electronics network environment. In cooperative game theory [18], we need players and coalitions as:

$$N = \{1, \dots, n\} \quad (4)$$

and

$$Coalitions = S \subseteq N \quad (5)$$

Then we need a characteristic functions. Actually the characteristics function game G by a pair (N, V) , Where,

$$V : 2^N \longrightarrow \mathbb{R} \quad (6)$$

Here Eq. (6) is the characteristics function which maps each coalitions $S \subseteq N$ real number $V(S)$. Then we can assume pair distribution as: 1. Efficiency, 2. Null Players, 3. Symmetry and 4. Additivity Then the cooperative shapely value will be $G = (N, V)$, the shapely value of a player $i \in N$ is given by

$$\phi_i(G) = \frac{1}{|N|!} \sum_{\pi \in \Pi N} \Delta_{\pi}^G(i) \quad (7)$$

The main objective of cooperative Shapley value is to measure the distribution of the power. In simple words, it is used to measure the voting game player's distribution. We can adopt cooperative Shapley value and rank and effective weight features in our proposed technique through this base. The cooperative Shapley value is very effective and efficient to find out the feature importance when several features for identification attack traffic. However, we also adopted this technique in this paper for effective selection and recommendation by using Shapley value as discussed in the following with a detailed explanation. The Cooperative Shapley value can be denoted by $(\phi_i(v))$,

$$\phi_i(v) = \sum_{k \subset N} \Delta_i(k) x \frac{|k|!(n - |k| - 1)!}{k!} \quad (8)$$

Similarly, in the Eq. (9) n is the number of players and sum over all subset K of N , where there no player i .

$$\Delta_i(k) = v(k \cup i) - v(k) \quad (9)$$

Likewise, Eq. (6) represent $\psi(ij)$

$$\psi(ij) = \begin{cases} 1, & I(f_j; class|f_i) > I(f_j; class) \\ 0, & \text{else} \end{cases} \quad (10)$$

$$\psi(ij) = \begin{cases} 1, & I(f_j; class|f_i) > I(f_j; class) \\ 2, & \text{else} \end{cases} \quad (11)$$

Similarly, to redefine the above equation definitions as

$$\Delta(k) = \begin{cases} 1, & I(k; class; f_i) \geq 0 \text{ and } \sum_{f_j \in k} \psi(ij) \geq \frac{|k|}{2} \\ 0, & \text{else} \end{cases} \quad (12)$$

However, the Coalitions of each players for each features are shown in Fig. 4.

Then the final result of shapely value are shown in Fig. 7.

B. Proposed Algorithm

This section presents our proposed effective feature selection technique named CoopGBFS to identify harmful traffic.

Value of coalition {1}:	84
Value of coalition {2}:	84
Value of coalition {2, 1}:	91
Value of coalition {3}:	91
Value of coalition {3, 1}:	91
Value of coalition {3, 2}:	91
Value of coalition {3, 2, 1}:	91
Value of coalition {4}:	91
Value of coalition {4, 1}:	91
Value of coalition {4, 2}:	91
Value of coalition {4, 2, 1}:	91
Value of coalition {4, 3}:	91
Value of coalition {4, 3, 1}:	91
Value of coalition {4, 3, 2}:	91
Value of coalition {4, 3, 2, 1}:	91
Value of coalition {5}:	92
Value of coalition {5, 1}:	92
Value of coalition {5, 2}:	92
Value of coalition {5, 2, 1}:	92
Value of coalition {5, 3}:	92
Value of coalition {5, 3, 1}:	92
Value of coalition {5, 3, 2}:	92
Value of coalition {5, 3, 2, 1}:	92

Fig. 4. Coalition 1.

Algorithm: FeatureRecommendationSelection(Δ):

```

1. Begin;
2.  $\Delta := load\_dataset()$ 
3.  $\Sigma := input\_features\_set()$ 
4. while true do
5.    $v := count\_features(\Sigma)$ 
6.    $\omega := calculate\_weight\_accuracy(\Delta, \Sigma)$ 
7.    $\Gamma := calculate\_coalition(\omega)$ 
8.    $\phi := apply\_cooperative\_shapley\_value(\Gamma)$ 
9.    $P := wrapper\_technique(\phi)$ 
10.   $\Phi := select\_effective\_features(P)$ 
11.  if check\_selected\_features( $\Phi$ ) then
12.     $display\_results(\Phi)$ 
13.    return
14.  end if
15.   $P' := remove\_unselected\_features(\Phi)$ 
16.  if  $P'$  is empty then
17.     $display\_results(\Phi)$ 
18.    return
19.  end if
20.   $\Sigma := P'$ 
21. end while

```

Fig. 5. Proposed CoopGBFS Algorithm A.

1) CoopGBFS Algorithm: In this subsection, we explained the proposed CoopGBFS algorithm-based cooperative game theory. The objective of the proposed algorithm is to select effective features for the identification of malicious attacks in IoT networks using a machine learning algorithm, as shown in Fig. 5 and 6. For the better understanding the proposed algorithm his two part “A” and “B” As discussed in the previous section, the proposed algorithm objective is to select the best features set from the given several features. For this objective, the proposed algorithm is divided into three phases, as shown in Fig. 8. In the first phase, the features set is supplied,

Algorithm: *FeatureRecommendationSelection(Δ):*

```

1. begin;
2. int count_features(set<string> features) {
3.   int count = 0;
4.   for (auto feature : features) {
5.     count++;
6.   }
7.   return count;
8. }
9. unordered_map<string, float> calculate_weight_accuracy
   (dataset  $\Delta$ , set<string>  $\Sigma$ ) {
10.  unordered_map<string, float>  $\omega$ ;
11.  for (auto feature :  $\Sigma$ ) {
12.    float sum = 0.0;
13.    for (auto data :  $\Delta$ ) {
14.      if (data.classification == 1) {
15.        sum += data.features[feature];
16.      }
17.    }
18.    float accuracy = sum / count_malicious( $\Delta$ );
19.     $\omega$ [feature] = accuracy;
20.  }
21.  return  $\omega$ ;
22. }

```

Fig. 6. Proposed CoopGBFS Algorithm B.

The Shapley value of player 1 is: 0.166666666666667
 The Shapley value of player 2 is: 0.166666666666667
 The Shapley value of player 3 is: 1.33333333333333
 The Shapley value of player 4 is: 1.33333333333333
 The Shapley value of player 5 is: 1.66666666666667
 The Shapley value of player 6 is: 2.16666666666667
 The Shapley value of player 7 is: 8.16666666666667

Fig. 7. Cooperative Shapley Values.

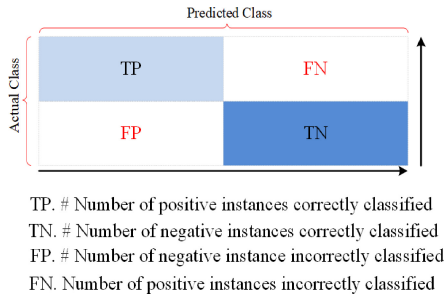


Fig. 8. Confusion Matrix.

and then it counted to calculate the number of features is available. After calculation, the weight accuracy is calculated. Here, accuracy is calculated after applying the correlation coefficient technique, and the weight is indicated the accuracy of a feature. After calculating accuracy, the algorithm transfer control to the second phase, in which the coalition of counted accuracy feature is calculated. As discussed in the previous section, coalitions are essential in cooperative game theory, especially in the cooperative Shapley value technique. Then cooperative Shapley value is applied based on coalition values of features. Finally, in the third phase, the wrapper technique is used based on the cooperative Shapley value to rank the best feature for the identification of malicious attacks in IoT consumer electronics network environments. The proposed algorithm will select effective features and remove the redundant features. Similarly, if the values of selected features are

not high enough, then the algorithm control transfer to remove otherwise forward to Swrapper step.

Let Δ be the dataset, and let Σ be the set of input features. We begin by loading the dataset Δ and initializing Σ :

$$\Delta = \text{load_dataset}() \quad (13)$$

$$\Sigma = \text{input_features_set}() \quad (14)$$

We then enter a loop to iteratively select a subset of features:

$$\text{while_true_do} \quad (15)$$

Within each iteration of the loop, we first calculate the number of features in the current set Σ :

$$v = \text{count_features}(\Sigma) \quad (16)$$

We then calculate the weight accuracy of each feature in Σ with respect to the dataset Δ :

$$\omega = \text{calculate_weight_accuracy}(\Delta, \Sigma) \quad (17)$$

Using the weights ω , we compute the Shapley value of each feature, taking into account the cooperative effect of features in Σ :

$$\Gamma = \text{calculate_coalition}(\omega) \quad (18)$$

$$\phi = \text{apply_cooperative_shapley_value}(\Gamma) \quad (19)$$

We then apply a wrapper technique to select a subset of features ρ from the original set Σ :

$\rho = \text{wrapper_technique}()$ $\rho = \text{wrapper_technique}(\Phi)$. We select the most effective features from ρ :

$\phi = \text{select_effective_features}(\rho)$. If the selected features in Φ meet some criterion, we display the results and terminate the algorithm: **if** check_selected_features(Φ) **then** display_results(Φ) **return** end **if**. If the selected features do not meet the criterion, we remove the unselected features from ρ to obtain a new set of input features ρ' :

$$\rho' = \text{remove_unselected_features}(\Phi) \quad (20)$$

If ρ' is empty, we display the current selected features and terminate the algorithm: **If** ρ' is empty **then** display_results(Φ) **then** **return** and **end** **if** statements. Finally, we update Σ with ρ' and continue to the next iteration: $\Sigma = \rho'$ **end while** $\Sigma = \rho'$ **statement**.

IV. EVALUATION METHODOLOGY

In this section, the detailed evaluation methodology is discussed, and the dataset used to evaluate our proposed technique is examined as follows subsection. Firstly, the dataset is discussed, and then later performance evaluation metrics are discussed.

2) *Data Set*: This subsection includes the dataset that is conducted for this study. For this objective, a Bot-IoT dataset is utilized for the proposed approach evaluation. In 2018, Koroniotis et al. [17] and Van der Elzen et al. [19] developed the Bot-IoT dataset to identify Bot-IoT malicious traffic flow in IoT networks. However, the dataset that was produced comprises IoT traffic, Bot-IoT traffic, regular traffic, and several attack traffic types that are frequently utilised in botnet assaults

in IoT networks. Furthermore, the realistic testbed, which includes a labelled feature set, is used for accurate trace development. They initially extracted a small collection of features for the machine learning technique's successful performance outcomes, but later expanded and included more features for the precise classification of Bot-IoT traffic flow. Attacks traffic, categories, and subcategories, among other things, were labelled. They divided the Internet of Things testbed into three sections, including features extraction, network platform, and forensics analytics. They integrated five IoT devices for the IoT data in more detail. For instance, an IoT device that generates weather data, such as the current temperature, humidity, air pressure, etc., is utilised to provide weather information. In a similar vein, the second Internet of Things (IoT) gadget they employed for their research is an IoT-based refrigerator that automatically generates data on the interior temperature and regulates the temperature. Pseudorandom general signal-based lights are the third special IoT gadget that they used for their research. These lights work off of movement. Based on motion, the smart bulb turns on automatically when there is movement, and it stays off when there is no movement. The smart IoT door is the fourth one, and the smart thermostat is the fifth. This gadget is used to automatically manage and regulate the indoor temperature of the house.

3) *Performance Measurement*: For the identification of performance analysis of the proposed model, performance analysis is essential. For this objective, numerous researchers in the research community applied confusion metrics, particularly for machine learning (ML) model performance measurement. Therefore, confusion metrics are the critical metrics for the performance measurement and based on confusion metrics. In this paper, we also adopted this technique to find out the performance analysis of our proposed method. The graphical representation of confusion metrics is shown in Fig. 8. However, the graph includes rows and columns. Rows represent the class samples, and column represents the classified samples of a class. The most important and widely used confusion metrics that numerous researchers used for their proposed technique performance evaluation are discussed below in detail. For our proposed model performance evaluation, we adopted the following confusion metrics as follows:

The above showing metrics in the figure are widely applied for the ML model performance measurement. Based on the above-explained metrics, several different metrics can be made for better performance analysis. It is significant to note that effective machine learning (ML) classifiers always minimize the FP metrics value and FN metrics values for effective performance results. However, the metrics that we selected for this study are explained below:

- **Accuracy (ACC)**: In cyber-attack identification or detection, the accuracy metric is the samples that are correctly identified in overall classified samples. However, the detailed explanation of the accuracy can be defined mathematically below. Therefore, we adopted this accuracy metric for our proposed model to measure the performance of the model. Furthermore, using this metric

effectiveness of the proposed model can be easily identified.

$$Accuracy = \frac{(TP + TN)}{(TP + TN + FP + FN)} \quad (21)$$

- **Precision**: the precision metric can be defined as Class X correctly classified sample traffic in all those classified in Class X. In a simple way, the true positive metrics divide by true positive and false positive. The mathematical equation that we used for this study is shown below.

$$Precision = \frac{TP}{(TP + FP)} \quad (22)$$

- **Sensitivity**: This metric can be defined as the correctly identified sample divide by the overall sample. This metric is also known as the recall metric. The mathematical equation that we used in this study is given below.

$$Sensitivity = \frac{TP}{(TP + FN)} \quad (23)$$

- **Specificity**: This metric is a very simple metric that several researchers use for their performance analysis. This metric indicates that a metric that can identify the negative results is called specificity metric. The mathematical equation is given below.

$$Specificity = \frac{TN}{(FP + TN)} \quad (24)$$

Therefore, we applied the above-discussed metrics for our proposed model performance measurements. The detailed results analysis of the proposed model is presented in the next section.

A. Results and Analysis

This section covered the in-depth findings and analysis of our suggested method, CoopGBFS, for selecting appropriate features to detect cyberattack traffic flows in IoT consumer electronics network environments. to use machine learning (ML) techniques to solve the issue of effective feature selection and determine which feature has discriminative capacity to appropriately identify cyber-attacks traffic in IoT networks. In order to identify cyberattack traffic in the IoT network, we presented the efficient feature selection method CoopGBFS and efficiently picked five different features. Four well-known machine learning (ML) algorithms Support Vector Machine (SVM), Decision Tree (C4.5), Naive Bayes, and Random Forest (RF) are used for the proposed technique, which makes use of the CoopGBFS assessment Bot-IoT dataset. With the help of the useful features set suggested by our feature selection approach and the corresponding performance assessment metrics of accuracy, precision, sensitivity, and specificity metrics, the four applied ML algorithms show promising results for identifying cyber-attacks in IoT networks. However, the outcomes of all used ML algorithms are promising. Similar to this, using the chosen feature set, the decision tree (C4.5) algorithm produces high accuracy results for detecting cyberattacks in IoT networks.

Likewise, Random Forest and Naïve Bayes accuracy results are also very effective using selected features. However, the

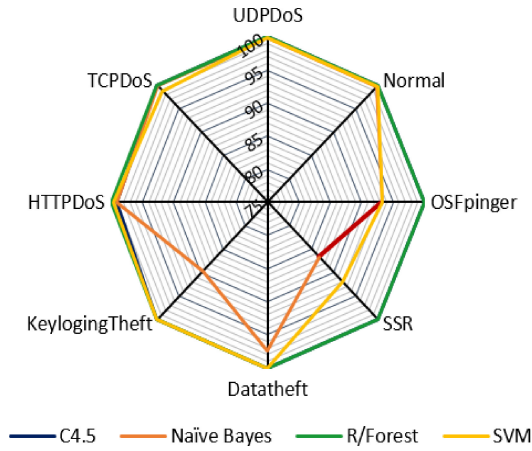


Fig. 9. Accuracy Results.

Random Forest ML algorithm is very effective compared to the Naïve Bayes algorithms accuracy result. Similarly, all the traffic flows are effectively identified using selected features. However, OSFpinger and SSR traffic are a bit slow identified compared to other flow traffic of IoT networks. But overall identification accuracy is very promising using ML algorithms and selected feature sets, as shown in Fig. 9. However, all the attacks and normal traffic flows are accurately classified, but only OSFpinger are a bit slow classified compared to other traffic flows, as shown in Fig. 9.

Similarly, in Fig. 10, the precision results of the applied ML algorithms are shown. The figure shows that UDPDoS attacks and Normal traffics are effectively identified compared to other traffic flows. But it is also clear that SSR is a bit slow identified than UDPDoS and Normal traffic flows with respective precision metrics. However, the SVM algorithm performance result using selected features set for precision is not very promising compared to other applied ML algorithms. But it is also seen that the SVM algorithms performance results are very promising for Normal and UDPDoS attacks, respectively 99% and 100%. However, the lowest precision result that the SVM gives is OSFpinger traffic flow. Naïve Bayes algorithm performance results are also very effective. However, for Data theft and Keylogging Theft attacks, Its performance result is not effective compared to other traffic flows. Though, the overall precision results for the identification of cyber-attacks are effective. But decision tree (C4.5) ML algorithm gives a promising performance result compared to other machine learning algorithms, as shown in Fig. 10.

Likewise, the SVM algorithms give very poor sensitivity results for sensitivity results compared to other ML algorithms, as shown in Fig. 11. But on another side, the SVM algorithm gives very effective sensitivity results for UDPDoS and SSR attacks traffic flows. However, the overall sensitivity results of SVM are not effective compared to other ML algorithms. In IoT consumer electronics network contexts, the decision tree C4.5 and Random Forest machine learning algorithms provide effective sensitivity results for identifying cyber-attacks. In contrast to other network flows, UDPDoS, TCPDoS, and SSR assaults traffic is properly segmented. On the other hand, as shown in Fig. 11, sensitivity results for detecting

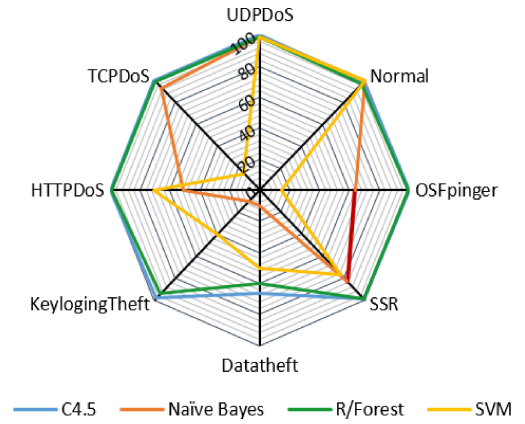


Fig. 10. Precision Results.

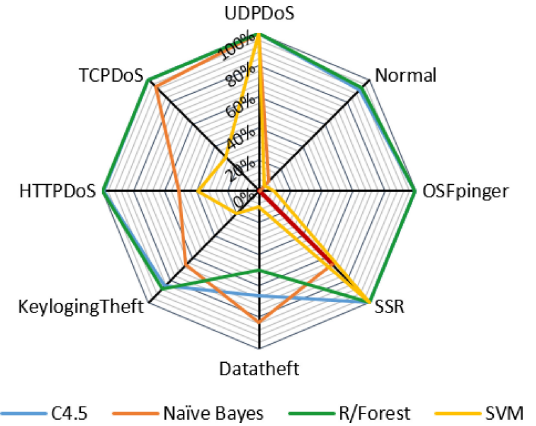


Fig. 11. Sensitivity Results.

cyber-attacks in the IoT consumer electronics network environment are effective in comparison. Similarly, the specificity findings of the used ML algorithms are shown in Fig. 12. Once more, the figure demonstrates that each ML method achieves distinct performance results that are effective. Even while the results of all applied ML algorithms are encouraging, the decision tree C4.5 ML algorithm produces the most encouraging outcomes when compared to other applied ML algorithms in terms of specificity metrics. In-depth classification of IoT technology traffic flows includes both attacks and regular traffic. SSR attacks, however, are yielding somewhat subpar findings in comparison to other traffic flow classification outcomes. Finally, Fig. 12 refSpecificity Results shows that our suggested approach successfully chooses robust features for detecting cyber-attacks in IoT networks.

V. CONCLUSION

To address security recommendation problem for cyber-attacks detection in the 5G IoT consumer electronics network using ML algorithms with federated learning dataset. A new framework model named CoopGBFS based on cooperative game theory and correlations with algorithm accuracy is proposed. Firstly, an Automatic Data set Generator (ADG) method is proposed and then based on proposed method an algorithm named ADG is designed and developed for the effective features set data set. Then based on proposed

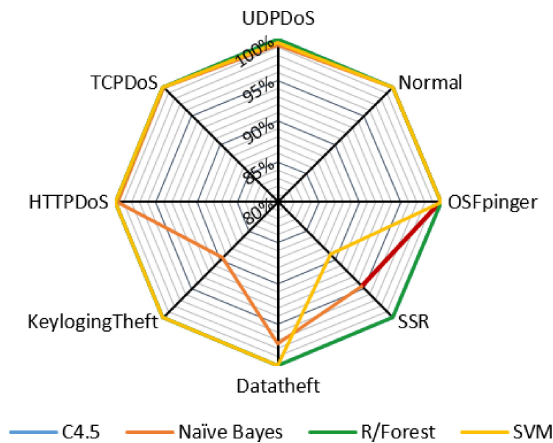


Fig. 12. Specificity Results.

ADG algorithm we proposed a new recommendation model CoopGBFS, based on cooperative game theory, federated learning and correlations with algorithm accuracy to address the problem. Finally, based on proposed CoopGBFS method an algorithm named CoopGBFS is developed and designed to select and recommend effective features set for 5G IoT network security. Based on the analysis of the experimental results, it is apparent that the proposed methodology is effective in detecting and recommending a set of features that are capable of effectively identifying cyber-attack traffic in 5G IoT networks. The proposed model demonstrates a higher level of depth and effectiveness, as it incorporates carefully selected features. The chosen features demonstrate efficacy and provide sufficient information for the detection of cyber-attacks. Likewise, all of the machine learning algorithms employed in the study yielded encouraging outcomes in the realm of cyber-attack detection.

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